NLP Coursework

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Repository: `https://github.com/freddienunn/NLP-Coursework`

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1 Exercise 1: Critical Paper Review

Paper reviewed: Pérez-Almendros et al., "Don't Patronize Me! An Annotated Dataset with Patronizing and Condescending Language towards Vulnerable Communities", COLING 2020.

Q1. Primary Contributions

The paper's main contribution is the *Don't Patronize Me!* (DPM) dataset: 10,637 paragraphs from the News on Web corpus, annotated for Patronising and Condescending Language (PCL) towards vulnerable communities (e.g. refugees, homeless people, women). Before this, no large-scale dataset existed targeting PCL in news media specifically. Alongside the dataset, the authors introduce a taxonomy of seven fine-grained PCL categories (e.g. *Unbalanced power relations*, *Presupposition*, *Compassion*) which enables span-level annotation beyond simple binary labels. Finally, the paper provides baseline results across a range of models — from TF-IDF SVMs to fine-tuned BERT and RoBERTa variants — establishing a performance floor for both binary detection and multi-label category classification.

Q2. Technical Strengths

The annotation design is a clear strength. Rather than asking for a simple binary label, annotators used a three-way scale (0/1/2), combined into a 5-point composite score, with disagreements referred to a third annotator. Reporting IAA both with and without borderline cases (41% vs. 61% Cohen's Kappa) is honest about task difficulty. The error analysis is also notably thorough — the paper explains *why* models fail on specific examples (e.g. that detecting *Presupposition* requires world knowledge the model lacks), rather than just reporting aggregate numbers. The breadth of baselines, covering everything from BoW SVMs to BERT-large, also gives a useful picture of where added model complexity helps.

Q3. Key Weaknesses

The most significant weakness is that class imbalance is acknowledged (roughly 9% positive examples) but never addressed in the experiments — no oversampling, class-weighted loss, or threshold tuning is applied. This seems like a major gap given how directly it affects practical usability of the dataset. The binarisation threshold (treating composite scores 2–4 as positive) is also unjustified; grouping genuinely ambiguous cases (score 2) with clear positives (score 4) likely introduces noise, and no ablation is provided. Finally, the claim that BERT-large underperforms due to overfitting is asserted without supporting evidence such as learning curves or train/val loss plots, making it speculative.

2 Exercise 2: Exploratory Data Analysis

Technique 1: Class Distribution and Token Length Distribution

(Appendix: Basic Statistical Profiling)

Figure 1 reveals two structural properties of the training data. The dataset is severely class-imbalanced: 7,581 examples (90.5%) carry the No PCL label and only 794 (9.5%) carry the PCL label, a ratio of roughly 9.55:1. PCL examples are also marginally longer on average (mean 53.5 vs. 48.2 whitespace tokens; 95th percentile 114 vs. 101 tokens), though the distributions overlap substantially.

The imbalance is the more consequential finding. A model trained with standard cross-entropy and no correction will tend to predict the majority class, reaching ∼90% accuracy while failing on almost all PCL examples — which is why F1 on the positive class is a more appropriate metric than accuracy. This motivates using a class-weighted or focal loss to push the model to take the minority class seriously. The length distribution, with a 95th percentile of 114 tokens, supports setting `max_length = 128` for the tokeniser, which covers almost all examples without excessive padding overhead.

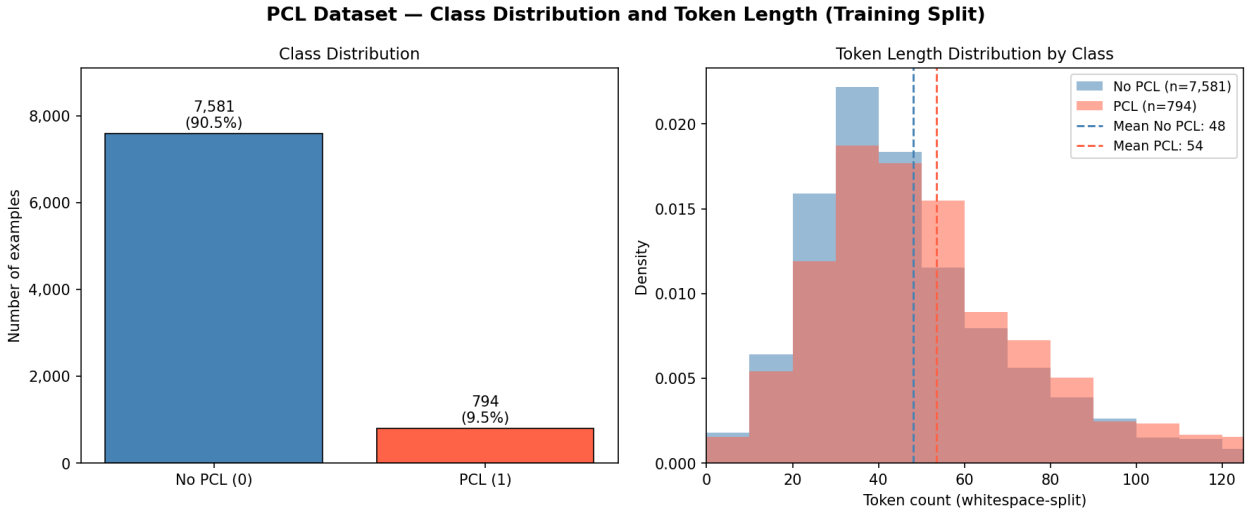

Figure 1: Left: class distribution of the training split. Right: overlapping density-normalised histograms of whitespace-tokenised text lengths for each class, with dashed vertical lines at each class mean.

Technique 2: N-gram Analysis — Top Bigrams by Class

(Appendix: Lexical Analysis)

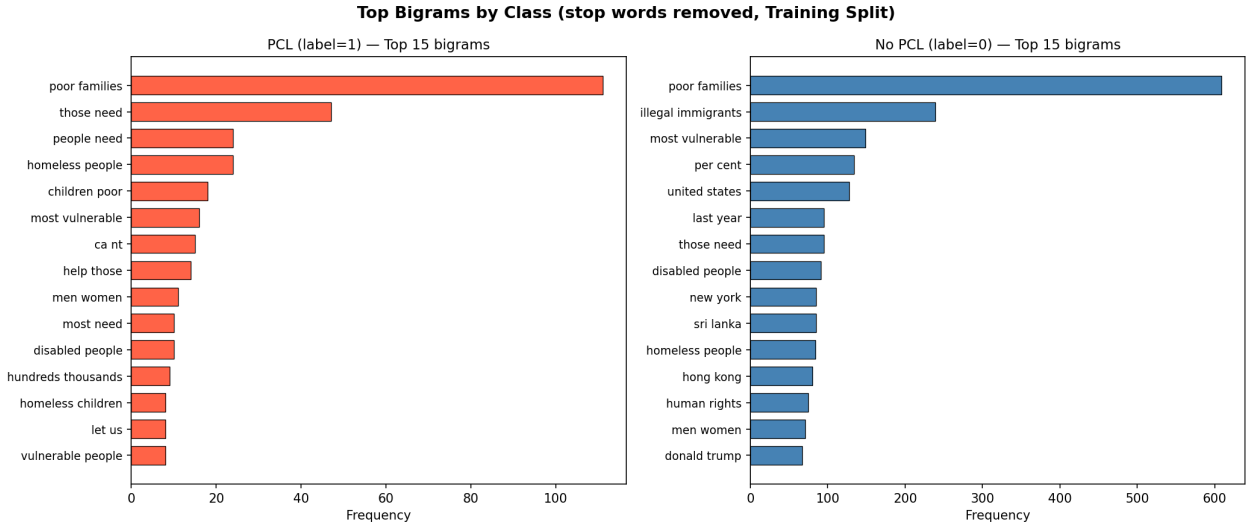

Figure 2: Top 15 most frequent bigrams (stop words removed) in PCL examples (left) and No PCL examples (right) of the training split.

Figure 2 compares the most frequent bigrams in each class after filtering stop words. While both classes share *poor families* as their top bigram, the subsequent distributions differ noticeably. PCL examples are dominated by relational, needs-oriented phrases — *those need*, *people need*, *help those* — that position the vulnerable group as passive recipients of concern. No PCL examples skew toward neutral, factual language: geographic references (*united states*, *new york*, *sri lanka*), statistics (*per cent*), and descriptive labels (*illegal immigrants*, *last year*).

The key takeaway is that the two classes are *not* separable by topic keywords alone — both discuss the same vulnerable communities using many of the same content words. The difference lies in framing and stance rather than subject matter, which rules out simple bag-of-words approaches and motivates using a contextual

model like RoBERTa that can represent sentence-level structure. It also suggests that features based on topic alone (e.g. the presence of *homeless*) would be unreliable, since such words appear frequently in both classes.

3 Exercise 3: Proposed Approach

Proposed Approach

The proposed model is a multi-task RoBERTa-base with two classification heads: a primary binary PCL head trained with focal loss, and an auxiliary 7-category head trained with BCE loss, jointly optimised on the full training set, with a decision threshold tuned on the dev set.

Architecture

The backbone is `roberta-base` (HuggingFace Transformers), a 125M-parameter encoder pre-trained with masked language modelling using dynamic masking and no next-sentence prediction objective. It produces a 768-dimensional `[CLS]` representation well-suited to sentence-level classification.

This representation is passed through a dropout layer ($p = 0.1$) into two parallel linear heads:

- Binary head ($768 \to 2$): predicts PCL vs. No-PCL. This is the only head used at inference time.
- Category head ($768 \to 7$): predicts which of the 7 fine-grained PCL categories are present (multi-label). This head is used only during training as an auxiliary regulariser.

The combined loss is: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda \cdot \mathcal{L}_{\text{BCE}}$, where $\lambda = 0.5$.

Training Methodology

Five deviations from the baseline are made, each motivated by the EDA:

1. **Focal loss.** The EDA revealed a 9.55:1 class imbalance. Rather than static class weights, focal loss [**?**] dynamically down-weights well-classified examples: $\text{FL}(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t)$, with $\gamma = 2.0$ and $\alpha = [1.0, 9.55]$. This concentrates learning on hard, ambiguous examples near the decision boundary rather than the easy majority.
2. **Multi-task learning with auxiliary category prediction.** The dataset provides 7 fine-grained PCL category labels for every example. Training the shared encoder to simultaneously predict these via a BCE head encourages richer, more structured PCL representations. At inference, only the binary head is used; the category head acts purely as a training-time regulariser.
3. **More training epochs.** The baseline trains for 1 epoch. Here the model is trained for 5 epochs, saving the best checkpoint by dev F1.
4. **Learning-rate schedule with warmup.** A linear warmup over 10% of training steps followed by linear decay stabilises early training. The baseline uses no scheduler.
5. **Decision threshold optimisation.** After training, τ is swept over $\{0.05, 0.10, \ldots, 0.95\}$ on the dev set and the value maximising positive-class F1 is selected, directly targeting the evaluation metric.

The full hyperparameter configuration is listed in Table 1.

Rationale and Expected Outcome

RoBERTa-base was chosen as the backbone because it is a well-understood encoder for sentence classification and a natural extension of the BERT baselines in the paper. It was pre-trained on a larger corpus than BERT with improved methodology (dynamic masking, no NSP, larger batches), giving better general representations while still being tractable to fine-tune on a dataset of this size.

Focal loss is preferred over static class weights because it is dynamic: the modulating factor $(1 - p_t)^\gamma$ automatically reduces the contribution of easy examples and focuses training on hard cases near the boundary.

Hyperparameter	Value
Backbone	`roberta-base`
Max sequence length	128
Primary loss	Focal loss, $\gamma = 2.0$, $\alpha = [1.0,\ 9.55]$
Auxiliary loss	BCEWithLogitsLoss (7 PCL categories)
Auxiliary weight (λ)	0.5
Training epochs	5
Batch size	16
Learning rate	2×10^{-5}
Warmup	10% of total steps (linear)
LR schedule	Linear decay after warmup
Optimiser	AdamW, weight_decay = 0.01
Training examples	8,375 (full set, no downsampling)

Table 1: Hyperparameter configuration for the proposed multi-task RoBERTa-base model.

For PCL detection, where the distinction between condescending framing and neutral reporting is often subtle, this seems particularly well-suited.

The motivation for multi-task learning is that the 7 PCL categories capture distinct linguistic mechanisms — e.g. *Unbalanced power relations* vs. *Shallow solution* vs. *Metaphor*. Training the encoder to distinguish these forces it to learn richer representations than a single-task model would, acting as an inductive bias toward understanding *how* condescension works rather than just that it is present. The auxiliary head is discarded at inference, so this adds no computational cost.

Threshold optimisation is included because focal loss with class-specific α weights shifts the output probability distribution away from a well-calibrated 0.5 threshold, so grid-searching on the dev set is a simple way to close that gap.

Together, each component addresses a specific weakness of the baseline, and a dev F1 in the range 0.55–0.65 is expected, compared to the baseline's 0.48.

4 Exercise 5: Predictions and Error Analysis

Exercise 5.1 — Dev and Test Predictions

Dev predictions (`dev.txt`) and test predictions (`test.txt`) were generated by the multi-task RoBERTa-base model described in Exercise 3, with a decision threshold of $\tau = 0.50$ selected by grid search on the dev set. The resulting dev F1 (positive class) is **0.6196**.

Exercise 5.2 — Local Evaluation

Part A: Error Analysis

The best model (multi-task RoBERTa-base, 5 epochs, focal loss + auxiliary category head, $\tau = 0.5$; dev F1 = 0.6196) and the spec baseline (single-task RoBERTa-base, 1 epoch, standard cross-entropy, 2:1 negative downsampling, $\tau = 0.5$; dev F1 = 0.5073) were compared on all 2,094 dev examples. Each example was assigned to one of four mutually exclusive outcomes.

The best model improves on the baseline in 160 cases while regressing on only 41 (a 14.6× improvement-to-regression ratio), confirming that the overall F1 gain is genuine and broadly distributed rather than concentrated on a few examples.

Only best model correct. The 160 cases where the best model succeeds but the baseline fails are dominated by examples annotated with *Presupposition*, *Compassion*, and *Authority voice* categories — patterns the 1-epoch CE model misses entirely:

Scenario	Count	%
Both correct	1783	85.1
Both wrong	110	5.3
Only best model correct	160	7.6
Only baseline correct	41	2.0
Total	2094	100.0

Table 2: Four-bucket comparison of the best model and the baseline on the 2,094 dev examples.

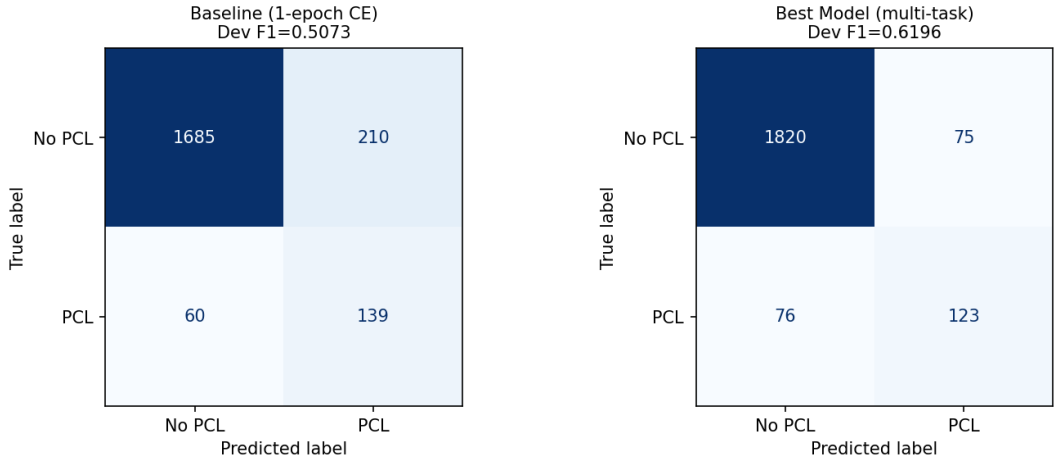

Figure 3: Confusion matrices for the baseline (left) and the best model (right) on the dev set. The best model substantially increases true positives while keeping false positives comparable.

- *"Hollywood star Leo Di Caprio urges help for reuniting immigrant children with their families"* — gold=1, category: Unbalanced power relations. The construction "[celebrity] urges help for [vulnerable group]" positions the community as passive objects of advocacy, a structural pattern the multi-task encoder detects.
- *"Rather sad. Good set of pictures that tells a tale of survival, subsistence living, and hopeless nature of life…"* — gold=1, categories: Presupposition, Metaphor, Compassion. The framing of poverty as immutable and "hopeless" is an implicit presupposition that longer focal-loss training brings into the model's representations.

Only baseline correct / both wrong. The 41 regressions are mostly positive examples the best model under-predicts: e.g. *"Our country is in need of serious change. We can continue to celebrate women's day, throw massive budgets at events…"* (category: Shallow solution), where sarcastic dismissal of superficial gestures is subtle. The 110 *both wrong* cases are uniformly false negatives, with *Presupposition* heavily represented — e.g. *"Many refugees don't want to be resettled anywhere, let alone in the US"*, where PCL is embedded in a clause-level assumption rather than explicit condescending phrasing.

Part B: Ablation Study

To isolate the contribution of the auxiliary category head, a single ablation was run: the same training setup (5 epochs, focal loss, full training set) but with the category head removed and only the binary PCL head trained. The optimal threshold for this ablation is $\tau = 0.65$, compared to $\tau = 0.50$ for the full model, reflecting a more conservative output distribution.

The ablation already achieves a substantial F1 of 0.6000, demonstrating that five epochs of focal-loss training alone accounts for most of the gain over the baseline (+0.0927 F1). The category head adds a further +0.0196

Model	F1	Precision	Recall
Baseline (1-epoch CE, 2:1 downsample)	0.5073	0.3983	0.6985
Ablation (5-epoch focal, no cat. head)	0.6000	0.6954	0.5276
Best model (5-epoch focal + cat. head)	0.6196	0.6212	0.6181

Table 3: Dev-set F1, precision, and recall for all three systems. The ablation isolates the effect of the auxiliary category head.

Scenario	Count	%
Both correct	1917	91.5
Both wrong	114	5.4
Only best model correct	26	1.2
Only ablation correct	37	1.8
Total	2094	100.0

Table 4: Four-bucket comparison of the best model and the ablation (no category head) on the 2,094 dev examples.

F1, but the nature of that improvement is more nuanced than the headline number suggests.

The key difference is in the precision/recall trade-off. The ablation is already very precise (P=0.6954) but under-predicts (R=0.5276): focal loss suppresses false positives effectively, but without the regularisation signal from the category head, the encoder is reluctant to commit to borderline PCL predictions. The category head, by forcing the encoder to represent the seven fine-grained PCL mechanisms simultaneously, pushes it to make more PCL predictions at a given confidence level. This closes the recall gap substantially (0.6181 vs. 0.5276) at a slight precision cost (0.6212 vs. 0.6954), and the net effect is a more balanced F1.

At the individual-example level the margin is narrow: the best model wins 26 cases the ablation misses, while losing 37 it gets right. The 26 best-model wins are mostly multi-category examples (e.g. Unbalanced power relations + Compassion, or Presupposition + Metaphor) where the encoder needs to simultaneously represent multiple PCL mechanisms. The 37 ablation wins are predominantly false positives that the full model avoids, consistent with its slightly lower precision. The 114 *both wrong* cases largely overlap with the best-vs-baseline hard set, confirming that they are genuinely ambiguous examples that neither architectural choice resolves.